0. Welcome, introduction round, organization and structure

- **Structure and contents of the course**

Lectures: 30 units of 45 min

1 Introduction
2 Economy and ecology of polymers
3 General properties and applications
4 Groups of polymers and classification
5 Polymer Chemistry
6 Amorphous (non-crystalline) polymers
7 Semicrystalline polymers
8 Polymer blends
9 Copolymers and block copolymers
10 Polymer composites and nanocomposites
11 Elastomers
12 Additives
13 Polymer processing and rheology
14 Polymer testing and characterization
15 Selection of polymers for certain applications

Exercises: 30 units of 45 min

(E1) Find polymers — Campus excursion
(E2) Identify polymers — Collection of samples
(E3) Student presentations — 15 + 5 min per student
(E4) Student presentations — 15 + 5 min per student
(E5) Student presentations — 15 + 5 min per student
(E6) Student presentations — 15 + 5 min per student
(LC1) Lab tour and introduction
(LC2) Mechanical testing
(LC3) Thermal and spectroscopic methods
(LC4) Microscopy
(LC5) Rheometry
(R) Repetition (weekly) — Repetition of main knowledge
Questions and answers (weekly)
Preparation for exam